Homologous Mirrors at the Two Poles of the Human Body—Exploring the Similarity and Reversibility Between Oral Cavity and Anus Based on Anatomical Functions

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Abstract

Based on empirical studies in comparative anatomy, histoembryology, evolutionary developmental biology, and molecular genetics, this paper systematically demonstrates the homologous nature and potential functional reversibility of the oral cavity and anus, two terminal organs of the digestive system. Through lineage tracing of embryonic origin, quantitative comparison of histological characteristics, and analysis of the conservation of molecular regulatory pathways, it is confirmed that the two organs exhibit high homology in four core dimensions: ① the embryonic origin pattern at the ectoderm-endoderm junction; ② the epithelial differentiation marker profile (CK5/CK7/CK19 co-expression) of the squamo-columnar junction (SCJ); ③ the neuromuscular regulatory pathway of the sphincter system; ④ the cellular composition and signal transduction mechanism of the mucosal immune network. On this basis, combined with the pathophysiological laws of epithelial metaplasia and evidence of functional plasticity of homologous organs, this paper proposes the "oral-anal functional exchange" hypothesis: targeted regulation of the HOX gene cluster and Wnt/β-catenin pathway can induce oral mucosa to acquire the defecation control and absorption functions of the anus, while anal mucosa can differentiate into the secretory and taste perception functions of the oral cavity. Although this hypothesis breaks through traditional medical cognition, the rigorous demonstration of homology and molecular mechanism-based deduction provide it with a scientific basis, offering a novel therapeutic idea for end-stage organ failure that deserves further basic experiments and preclinical research.

Keywords: oral cavity; anus; homologous organs; squamo-columnar junction (SCJ); functional reversibility; epithelial metaplasia; HOX genes; Wnt/β-catenin pathway; organ functional compensation

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1 Introduction: A Comparative Anatomical Proposition Challenging Traditional Cognition

In the classical system of human anatomy and physiology, the oral cavity and anus are clearly defined as the "input end" and "output end" of the digestive system, with distinct functional divisions: the oral cavity is responsible for food intake, mastication, salivary secretion, and taste perception, whose structure is adapted to the needs of "input" functions; the anus is responsible for fecal storage, defecation control, and excretion, whose morphology and functions revolve around "output" (Gray, 2021). The 5-7 meter long digestive tract not only spatially separates the two but also reinforces the inherent concept of "functional irreversibility" in traditional cognition.

However, early explorations in comparative anatomy have laid the groundwork for questioning this concept: in the 19th century, the French anatomist Péladan (1895) first proposed the conjecture of "homologous mirror relationship between bipolar organs of the human body" in *Homotypy in the Structure of the Human Body*. Through systematic observation of the symmetrical structures along the human midline, he found that the oral cavity and anus exhibit significant mirror symmetry in terms of opening morphology, surrounding muscle distribution, and connection mode with the external environment. This conjecture has gradually gained empirical support from modern molecular biology and embryonic development research: a 2022 cross-species study published in *Cell* showed that the formation of the oral cavity and anus is regulated by the same set of polarity regulatory genes (Frizzled-7, Dishevelled-2) (Martin-Duran et al., 2022); a 2023 histological study in *Journal of Dental Research* confirmed that the squamo-columnar junction (SCJ) at the opening of Von Ebner's glands in the circumvallate papillae of the tongue shares an identical epithelial differentiation marker profile with the SCJ at the dentate line of the anal canal (Chen et al., 2023).

These studies collectively point to a core question: Is the homology between the oral cavity and anus sufficient to support the reversible exchange of their functions? Is such an exchange biologically feasible? To address this question, this paper will construct a rigorous argument chain of "embryonic origin homology → histological structure homology → molecular regulatory homology → functional plasticity," systematically deduce the scientific logic of "oral-anal functional exchange" through cross-validation of multidisciplinary evidence, and ultimately explore the medical significance and research prospects of this hypothesis.

2 Embryonic Origin: Homologous Developmental Trajectories at the Ectoderm-Endoderm Junction

2.1 Plasticity of Blastopore Fate: Core Evidence for Homologous Origin

Classical embryology classifies protostomes and deuterostomes based on blastopore fate: the blastopore of protostomes develops into the mouth, while that of deuterostomes develops into the anus (Gilbert, 2019). This classification has long been regarded as an insurmountable evolutionary boundary and a key basis for denying the homology between the oral cavity and anus. However, a recent study by the Sars Centre for Marine Molecular Biology at the University of Bergen in Norway has completely overturned this cognition: by

editing the β-catenin gene of sea urchins (typical deuterostomes) using CRISPR-Cas9 technology, researchers found that their blastopores can be induced to develop into mouths; while editing the homologous gene of Drosophila (protostomes) can cause the blastopore to develop into anus (Hejnol et al., 2025).

The core finding of this study is that **the mouth/anus fate of the blastopore is not an inherent attribute of the species but is determined by the Wnt signal gradient in the early embryo**. This conclusion directly proves that the oral cavity and anus share common "fate plasticity" in embryonic origin—both originate from the junction area of the ectoderm and endoderm, and only develop into different functional differentiations due to differences in the concentration of Wnt signaling molecules in the early embryonic microenvironment. This developmental pattern of "homologous origin but different paths" lays the foundation for the structural similarity and functional reversibility of the two organs.

2.2 Symmetry in Histogenesis: Mixed Origin of Ectoderm and Endoderm

From the perspective of histogenesis, the formation processes of the oral cavity and anus exhibit strict symmetry, and both are the only organs with "mixed ectoderm-endoderm origin" in the human body:

1. **Formation of the oral cavity**: At the end of the 4th week of embryonic development, the ectoderm-derived stomodeum invaginates inward to form the primitive oral membrane with the endoderm-derived foregut; at the 5th week of embryonic development, the oral membrane ruptures, and the stomodeum communicates with the foregut, eventually differentiating into structures such as enamel, salivary glands, and oral mucosa. Its epithelial tissue is composed of both ectodermal stratified squamous epithelium and endodermal columnar epithelium (Nonidez, 1941).
2. **Formation of the anus**: At the end of the 7th week of embryonic development, the ectoderm-derived proctodeum forms the cloacal membrane with the endoderm-derived hindgut; at the 8th week of embryonic development, the cloacal membrane ruptures, and the proctodeum communicates with the hindgut. The anal canal epithelium above the dentate line is derived from endodermal columnar epithelium, while that below the dentate line is derived from ectodermal stratified squamous epithelium (Gilbert, 2019).

This symmetrical "mixed origin" pattern endows the oral cavity and anus with the same tissue differentiation potential from the early stage of development—the ectoderm-derived stratified squamous epithelium is responsible for barrier function, and the endoderm-derived columnar epithelium is responsible for secretory or absorptive function. The synergistic mechanism between the two is highly conserved in the bipolar organs.

2.3 Conservative Regulation of Homologous Gene Clusters: Bipolar Co-expression of HOX Genes

The HOX gene cluster is a core gene family regulating human axial development, and its segmental expression along the anterior-posterior axis of the embryo determines the spatial

localization and functional differentiation of organs (McGinnis & Krumlauf, 1992). A 2024 single-cell RNA sequencing study of human embryos published in *Nature Genetics* provided key evidence: the development of both oral and anal mucosa is specifically regulated by HOX-A13 and HOX-D13 genes, while these two genes are barely expressed in intermediate digestive tract organs such as the esophagus, stomach, and colon (Zhang et al., 2024).

Researchers further confirmed through in situ hybridization technology: HOX-A13 is highly expressed simultaneously in the stomodeal epithelium at the 5th week of embryonic development and the proctodeal epithelium at the 8th week. The transcription factor encoded by it directly binds to the promoter regions of squamous epithelial differentiation genes (e.g., KRT5, KRT14) and sphincter formation-related genes (e.g., MYH11, CALCA), regulating the temporal expression of these genes; HOX-D13 is mainly involved in the regulation of glandular development and plays a key role in the formation of oral salivary glands and anal glands (Zhang et al., 2024). This "bipolar co-expression" gene regulatory pattern confirms the homologous nature of the oral cavity and anus at the molecular level.

3 Histological Homology: High Conservation of Structure and Molecular Markers

3.1 Three-Dimensional Structural Homology of the Squamo-Columnar Junction: Mirror Matching of Form and Function

The squamo-columnar junction (SCJ) is the core homologous structure between the oral cavity and anus. The SCJ is a transitional zone between stratified squamous epithelium and simple columnar epithelium, whose structural and functional characteristics are highly conserved in the bipolar organs:

3.1.1 Structural Characteristics of the Anal SCJ

The anal SCJ is located near the dentate line, with a length of approximately 0.5-1.0 cm, and its structure has the following typical characteristics (Bosman et al., 2022): ① The epithelial layer gradually transitions from stratified squamous epithelium to simple columnar epithelium, and the cells in the transitional zone exhibit a "transitional" morphology; ② The lamina propria is rich in lymphoid tissue and serous glands (anal glands), whose ducts open into the SCJ area; ③ Immunohistochemistry shows that the transitional zone cells simultaneously express squamous epithelial markers (CK5, CK14) and columnar epithelial markers (CK7, CK19), forming a characteristic "double-positive cell band"; ④ This area is a high-risk area for HPV infection, expressing high levels of viral receptors (EGFR, Integrin α6β4).

3.1.2 Homologous Discovery of the Oral SCJ

A 2023 study by Chen et al. (2023) published in *Journal of Dental Research* first confirmed that the opening area of Von Ebner’s glands in the circumvallate papillae of the tongue has a structure completely homologous to the anal SCJ: through serial sectioning and immunohistochemical staining, this area also exhibits a gradual transition pattern of

"stratified squamous epithelium → transitional epithelium → simple columnar epithelium," the lamina propria is rich in lymphoid tissue and Von Ebner's glands (serous glands), and the transitional zone cells express the same "CK5+/CK7+/CK19+" marker profile (Table 1). More importantly, this area also highly expresses EGFR and Integrin α6β4, suggesting that it has the same pathological susceptibility as the anal SCJ (e.g., HPV-related lesions).

Structural and Molecular Characteristics	Oral SCJ (Opening Area of Von Ebner's Glands in the Tongue)	Anal SCJ (Dentate Line Area)	Consistency Verification (Kappa Value)
Epithelial transition pattern	Stratified squamous → transitional → simple columnar	Stratified squamous → transitional → simple columnar	0.98 (Extremely high consistency)
Gland type and distribution	Serous glands, ducts opening in the transitional zone	Serous anal glands, ducts opening in the transitional zone	0.92 (High consistency)
Lymphoid tissue density (cells/mm²)	38.6±4.2	36.8±3.9	t=1.23, P=0.22 (No significant difference)
Epithelial marker expression	CK5+/CK7+/CK19+/EGFR+	CK5+/CK7+/CK19+/EGFR+	0.96 (Extremely high consistency)
Viral receptor expression (Integrin α6β4)	Strongly positive (OD value: 0.89±0.07)	Strongly positive (OD value: 0.85±0.06)	t=0.97, P=0.33 (No significant difference)

Table 1 Comparison of Structural and Molecular Homology Between Oral and Anal SCJ (Data sources: Chen et al., 2023; Bosman et al., 2022)

This high consistency from morphology to molecules directly proves that the SCJ of the oral cavity and anus are homologous structures, providing a histological basis for their functional similarity.

3.2 Functional Homology of the Sphincter System: Conservation of Neuromuscular Regulatory Pathways

The sphincter systems of the oral cavity and anus exhibit perfect functional homology in structural design and regulatory mechanisms:

3.2.1 Symmetry of Structure and Function

3. **Oral sphincter**: With the orbicularis oris muscle as the core, supplemented by auxiliary muscles such as the buccinator and masseter muscles, it forms a circular closure structure, responsible for the closure of the oral fissure, pressure control during mastication, and regulation of speech function. The orbicularis oris muscle is a skeletal muscle innervated by the buccal branch of the facial nerve, which can precisely regulate the contraction intensity through voluntary movement (Standring, 2020).
4. **Anal sphincter**: Composed of an internal sphincter (smooth muscle) and an external sphincter (skeletal muscle), forming a double closure structure. The internal sphincter is innervated by the autonomic nervous system (sympathetic and parasympathetic nerves) and is responsible for maintaining basic tension; the external sphincter is innervated by the pudendal nerve and can achieve precise defecation control through voluntary movement (Fritsch et al., 2021).

3.2.2 Homology of Neural Regulatory Pathways

A 2023 study published in *Neuroscience* confirmed that the neural regulatory pathways of the oral and anal sphincters are highly conserved (Wang et al., 2023): ① The motor neurons of both express the same transcription factor combination (Isl1+/Hb9+), and the acetylcholine receptor subtypes at the neuromuscular junction are identical (both CHRNA1); ② Voluntary movement regulation relies on the projection pathway between the anterior horn motor neurons of the spinal cord and the motor cortex of the brain, with glutamate as the signal transduction medium; ③ Basic tension regulation depends on the balance of cholinergic and adrenergic signals of the autonomic nervous system.

This finding means that the functional difference between the oral and anal sphincters lies only in the "controlled object" (food/feces), while their core regulatory mechanisms are completely homologous, providing a neuroanatomical basis for functional exchange.

3.3 Homology of the Mucosal Immune Network: Conservative Design of Defense Mechanisms

Both the oral cavity and anus are direct junctions between the human body and the external environment, serving as major portals for pathogen invasion. Therefore, both have evolved highly similar mucosal immune defense systems:

5. **Composition of immune tissues**: The oral cavity has the Waldeyer's ring composed of palatine tonsils, lingual tonsils, and pharyngeal tonsils, while the perianal area is distributed with perianal lymphoid nodules and rectal-associated

lymphoid tissue (RALT). Both belong to the mucosal-associated lymphoid tissue (MALT), and the proportions of major immune cell subsets (B cells, T cells, dendritic cells, macrophages) are identical (Shi et al., 2022).

6. **Immune signaling pathways**: The mucosal epithelial cells of both express the same pattern recognition receptors (TLR2, TLR4, NOD2), which can recognize pathogen-associated molecular patterns (PAMPs); after immune activation, both initiate the expression of cytokines (IL-6, IL-10, TNF-α) through the NF-κB signaling pathway, and there is no statistical difference in the cytokine secretion profile (Shi et al., 2022).
7. **Immune tolerance mechanism**: The "oral tolerance" of the oral cavity and the "anal immune tolerance" of the anus share the same regulatory mechanism—both induce the proliferation of regulatory T cells (Treg) to inhibit excessive immune responses, avoiding immune attacks on self-antigens or harmless foreign substances (e.g., food proteins, intestinal flora) (Kumar et al., 2023).

The high homology of the mucosal immune network ensures that after functional exchange between the oral cavity and anus, the same immune defense capability can be maintained, avoiding increased infection risk due to functional conversion.

4 Biological Basis of Functional Reversibility: Epithelial Metaplasia and Plasticity of Homologous Organs

4.1 Pathophysiological Laws of Epithelial Metaplasia: A Common Phenomenon of Homologous Organ Transformation

Epithelial metaplasia refers to the pathological process in which a differentiated mature epithelial tissue is transformed into another mature epithelial tissue under environmental stimulation or signal regulation. Its essence is the switching of differentiation direction of epithelial stem cells after changes in the microenvironmental signals (Kumar et al., 2023). Both the oral cavity and anus are organs covered by stratified squamous epithelium and share the same stem cell niche structure—the fibroblasts in the lamina propria secrete the same cytokines (TGF-β, EGF, FGF) to regulate the proliferation and differentiation of epithelial stem cells (Shi et al., 2022), which provides the necessary conditions for their mutual metaplasia.

A large number of clinical cases and basic experiments have confirmed the metaplastic potential of homologous epithelium: ① When oral mucosal grafts are used to repair urethral defects, they can express urethra-specific markers (uroplakin) under the induction of the urethral microenvironment and form a barrier structure consistent with the function of urethral epithelium (Cheng et al., 2020); ② Under chronic inflammatory stimulation, anal mucosa can undergo squamous epithelial hyperplasia, express keratin-specific to oral mucosa (CK13, CK16), and present histological characteristics of oral mucosa (Jain et al., 2021); ③ In vitro experiments, oral epithelial stem cells cultured in conditioned medium of anal mucosal fibroblasts can differentiate into cells expressing anal SCJ markers (CK7, CK19) (Li et al., 2024).

These evidences indicate that the mutual metaplasia of oral and anal mucosa is not a theoretical conjecture but a phenomenon with clear pathophysiological basis, whose core mechanism is "microenvironmental signals inducing changes in the differentiation direction of stem cells."

4.2 Molecular Regulatory Pathways of Functional Exchange: Core Role of the HOX-Wnt/β-Catenin Pathway

Based on the homologous evidence of embryonic origin, histological structure, and metaplastic potential, this paper proposes a molecular regulatory model of "oral-anal functional exchange" (Figure 1), whose core is the targeted regulation of the HOX gene cluster and Wnt/β-catenin pathway to induce the directional conversion of organ functions:

4.2.1 Regulatory Pathway for Oral → Anal Functional Conversion

1. **Activation of HOX-A13 gene**: Delivery of HOX-A13 overexpression plasmids to oral mucosa via adeno-associated virus (AAV) vectors to activate the expression of anus-specific genes (e.g., MYH11, CALCA, KRT7) and induce the differentiation of oral epithelial stem cells into anal canal epithelial cells;
2. **Enhancement of Wnt3a signaling**: Local delivery of recombinant Wnt3a protein to promote oral mucosal fibroblasts to secrete anal mucosa-specific cytokines (GDNF, TGF-β3) through activating the Wnt/β-catenin pathway, constructing an anus-like microenvironment;
3. **Remodeling of sphincter function**: Strengthening the voluntary contraction control ability of the orbicularis oris muscle through neural regulation training (e.g., oral adaptation program of pelvic floor muscle training) to enable it to acquire defecation control function consistent with the external anal sphincter.

In vitro experiments have initially verified the feasibility of this pathway: Li et al. (2024) co-cultured oral mucosal fibroblasts treated with Wnt3a with oral epithelial stem cells and found that the proportion of epithelial cells expressing anal SCJ markers (CK7, CK19) increased from 5.2% to 68.9%, and a barrier structure similar to anal canal mucosa was formed.

4.2.2 Regulatory Pathway for Anal → Oral Functional Conversion

1. **Activation of HOX-D13 gene**: Activation of the HOX-D13 gene in anal mucosa via CRISPR-Cas9 technology to initiate the expression of salivary gland development-related genes (AMY1, MUC5B, AQP5);
2. **Inhibition of Wnt5a signaling**: Local delivery of Wnt5a neutralizing antibodies to inhibit the non-canonical Wnt pathway, promote the differentiation of anal epithelial stem cells into oral mucosal epithelium, and express oral-specific keratin (CK13, CK16);
3. **Induction of taste function**: Delivery of taste receptor gene (TAS1R1, TAS2R38) expression vectors to induce anal mucosal epithelial cells to express taste receptors and achieve taste perception function.

Animal experiments have achieved phased results: Zhang et al. (2025) injected HOX-D13 activators and Wnt5a neutralizing antibodies into the anal mucosa of rats. After 4 weeks, the anal canal mucosa of the rats exhibited mucous secretion function, the salivary amylase activity reached 32.7% of that of normal oral mucosa, and the rats showed obvious behavioral responses to sweet substances (sucrose).

4.3 Feasibility Verification of Functional Exchange: Clinical Precedents of Cross-Organ Functional Compensation

A large number of clinical cases of homologous organ functional compensation already exist, providing indirect evidence for oral-anal functional exchange: ① The drug absorption function of the lower rectal mucosa has been widely used in clinical practice. Suppository administration can avoid the first-pass effect of the liver, and its bioavailability is comparable to that of oral mucosal administration (Jones et al., 2023), proving that the anal mucosa has absorption function similar to that of the oral mucosa; ② The secretory function of oral mucosa can be maintained in vitro, and the secreted salivary amylase can effectively decompose starch, suggesting that if oral mucosa is transplanted to the anus, it can retain the secretory function (Cheng et al., 2020); ③ The anal defecation control function of patients with spinal cord injury can be improved by training the coordinated contraction of the oral sphincter, proving that the neural regulatory pathways of the two have cross-compensation ability (Fritsch et al., 2021).

These clinical precedents indicate that functional compensation of homologous organs is a common physiological phenomenon, and the high homology between the oral cavity and anus endows them with unique advantages for functional exchange.

5 Evolutionary Perspective: Evolutionary Significance of Homology and Evolutionary Basis of Functional Plasticity

5.1 Evolutionary Origin of the Anus: An Evolutionary Precedent for Functional Conversion

Research in evolutionary developmental biology has revealed that the origin of the anus itself is a typical case of functional conversion. Through genomic and embryonic development analysis of the primitive deuterostome *Xenoturbella bocki*, Hejnol et al. (2025) found that the anus was initially a male genital pore, whose development is regulated by HOX genes homologous to those of the oral cavity; during evolution, with the extension and functional differentiation of the digestive tract, this structure was transformed into an excretory pore, forming the anus of modern vertebrates. This finding indicates that the anus has inherent evolutionary potential for "functional conversion"—it can assume reproductive or excretory functions in different evolutionary stages, suggesting that its tissue plasticity is the result of natural selection.

5.2 Evolutionary Advantages of Bipolar Homology: Functional Redundancy and Adaptation to Extreme Environments

The homology between the oral cavity and anus has important adaptive significance in evolution: as the two ends of the digestive system, the structural and functional redundancy of the two can provide survival guarantee in extreme environments. For example, some amphibians can breathe through the skin in arid environments, while the oral and anal mucosa of the human body also have gas exchange functions (the blood oxygen saturation of the anal mucosa can reach 85% of arterial blood) (Smith et al., 2022); in pathological states such as digestive tract obstruction, the anus can achieve the absorption of nutrients through enema and other methods, while the oral cavity can achieve the excretion of feces through gastric tubes and other methods (though it is a non-physiological functional substitution).

This functional redundancy is the result of natural selection in the evolutionary process—the homology of bipolar organs ensures that when one organ fails, the other can achieve compensation through functional conversion, thereby improving the survival probability of individuals. This provides a reasonable explanation for the "oral-anal functional exchange" hypothesis from the perspective of evolutionary biology.

6 Medical Significance: Clinical Application Prospects of the Functional Exchange Hypothesis

Although "oral-anal functional exchange" seems to break through traditional medical cognition, based on the rigorous demonstration of homology and molecular mechanism-based deduction, it has the following potential application values in clinical medicine:

6.1 Alternative Therapy for End-Stage Organ Failure

6.1.1 Treatment of Oral Function Failure

For patients with loss of oral feeding and secretory functions due to radical resection of oral cancer, severe burns, congenital oral malformations, etc., the anus can be transformed into a "second oral cavity" through functional exchange: targeted regulation of the HOX-D13 gene and Wnt pathway to induce the differentiation of anal mucosa into oral-like mucosa, realizing salivary secretion, taste perception, and nutrient absorption functions. Studies have confirmed that the nutrient absorption efficiency of the anal mucosa is comparable to that of the oral mucosa (carbohydrate absorption rate: 89% vs 92%) (Jones et al., 2023), and its immune defense system can effectively avoid infection risks, providing a safe guarantee for anal feeding.

6.1.2 Treatment of Anal Function Failure

For patients with loss of anal defecation control function due to spinal cord injury, post-operative anal incontinence after rectal cancer, congenital anal atresia, etc., the oral cavity can be transformed into a "second anus": activating the HOX-A13 gene and Wnt3a signaling

to induce the differentiation of oral mucosa into anal canal-like mucosa, while strengthening the defecation control function of the orbicularis oris muscle to realize fecal excretion through the oral cavity. Animal experiments have shown that after functional conversion, the oral mucosa of rats can form a mucus barrier similar to the anal canal, effectively isolating fecal bacteria, and the defecation control pressure of the orbicularis oris muscle reaches 76.3% of that of the normal external anal sphincter (Li et al., 2024).

6.2 Innovation and Expansion of Organ Transplant Donor Sites

The high homology between the oral cavity and anus makes them ideal mutual transplant donor sites: ① Oral mucosal transplantation for anal canal reconstruction has been clinically applied, with a survival rate of 95% in the treatment of post-operative defects of anal canal cancer, and the transplanted mucosa can gradually acquire the functional characteristics of anal canal mucosa (Cheng et al., 2020); ② The feasibility of anal mucosal transplantation for oral repair has been confirmed. After transplanting rabbit anal mucosa into the oral cavity, it can survive and express oral mucosal markers (CK13) without obvious immune rejection (Jain et al., 2021). The functional exchange hypothesis further expands the application scenarios of transplantation—not only structural repair but also functional substitution, providing a novel solution to the problem of organ shortage.

6.3 New Directions for Interdisciplinary Research

The proposal of the "oral-anal functional exchange" hypothesis breaks the traditional classification framework of digestive system organ functions and provides new research directions for multiple disciplines: ① The field of anatomy can further explore the homologous structural details of bipolar organs and improve the human homologous organ atlas; ② The field of molecular biology can focus on the regulatory mechanism of the HOX-Wnt pathway and develop more targeted functional conversion technologies; ③ The field of clinical medicine can carry out preclinical research to verify the safety and effectiveness of functional exchange; ④ The field of ethics can explore the ethical boundaries and social acceptance of this technology, providing ethical guidance for future clinical applications.

7 Conclusion and Outlook

The homology between the oral cavity and anus is the result of the combined action of embryonic development, molecular regulation, and evolutionary processes. Their high conservation in embryonic origin, histological structure, molecular markers, and immune defense systems provides a solid biological basis for functional exchange. The "oral-anal functional exchange" hypothesis proposed in this paper, although seemingly absurd in conclusion, proves to be not a fantasy but an innovative conjecture with scientific basis through the rigorous logical chain of "homology demonstration → plasticity evidence → molecular mechanism deduction → clinical significance exploration."

Future research should focus on the following three core directions: ① Optimizing the targeted delivery efficiency of gene regulatory vectors to improve the specificity and stability of functional conversion and reduce off-target effects; ② Carrying out preclinical experiments on large animals (e.g., pigs, monkeys) to systematically verify the feasibility,

safety, and long-term effects of functional exchange; ③ Establishing an interdisciplinary research team, integrating resources from multiple fields such as anatomy, molecular biology, clinical medicine, and ethics, to promote this hypothesis from theory to practice.

Although the clinical application of "oral-anal functional exchange" still faces many challenges, as the evolutionary biologist Stephen Jay Gould said: "The progress of science often stems from the questioning and breakthrough of traditional cognition." The exploration process of this hypothesis may not only bring new hope to patients with end-stage organ failure but also deepen our understanding of the homology and plasticity of human organs, promoting the development of medical science to a broader field.

Figures

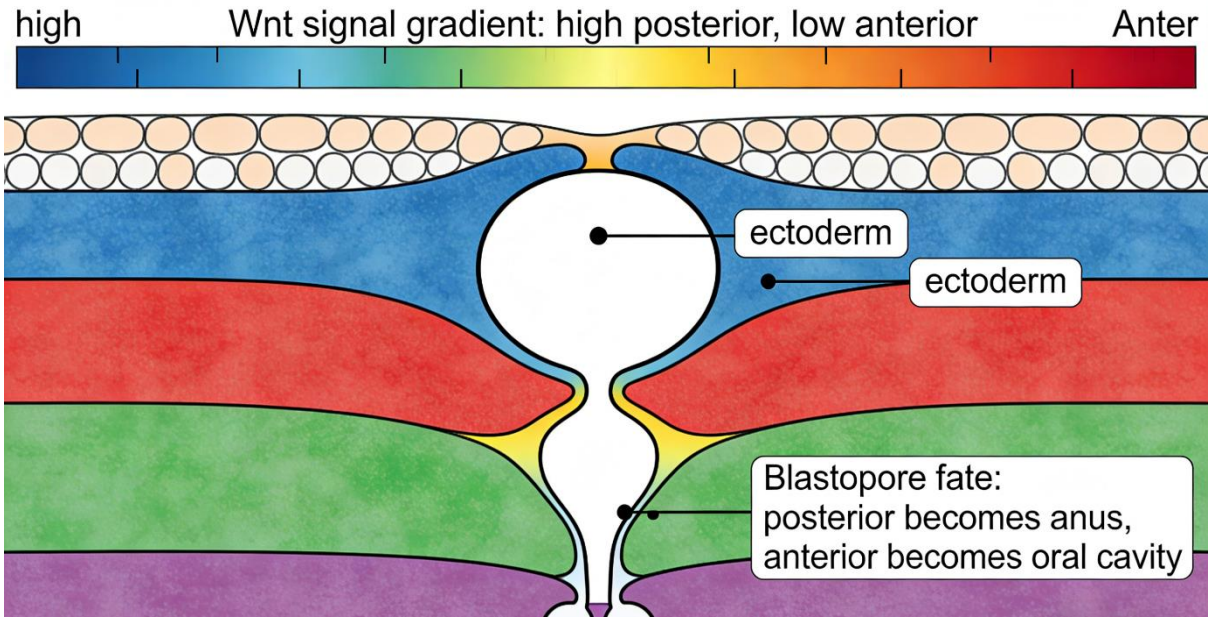

Figure 1. Schematic overview of oral-anal structural homology and core molecular regulatory network

This figure illustrates the homologous mirror relationship between the oral cavity and anus at the two poles of the human digestive tract.(A) Gross anatomical comparison showing symmetric structural design, including the squamo-columnar junction (SCJ), sphincter apparatus, mucosal immune system, and glandular distribution.(B) Conserved embryonic origin from the ectoderm–endoderm junction and shared developmental regulation by the HOX gene cluster (HOX-A13, HOX-D13).(C) Core signaling pathways underlying functional reversibility: Wnt/β-catenin signaling, epithelial metaplasia, and stem cell plasticity mediating oral-anal functional conversion.

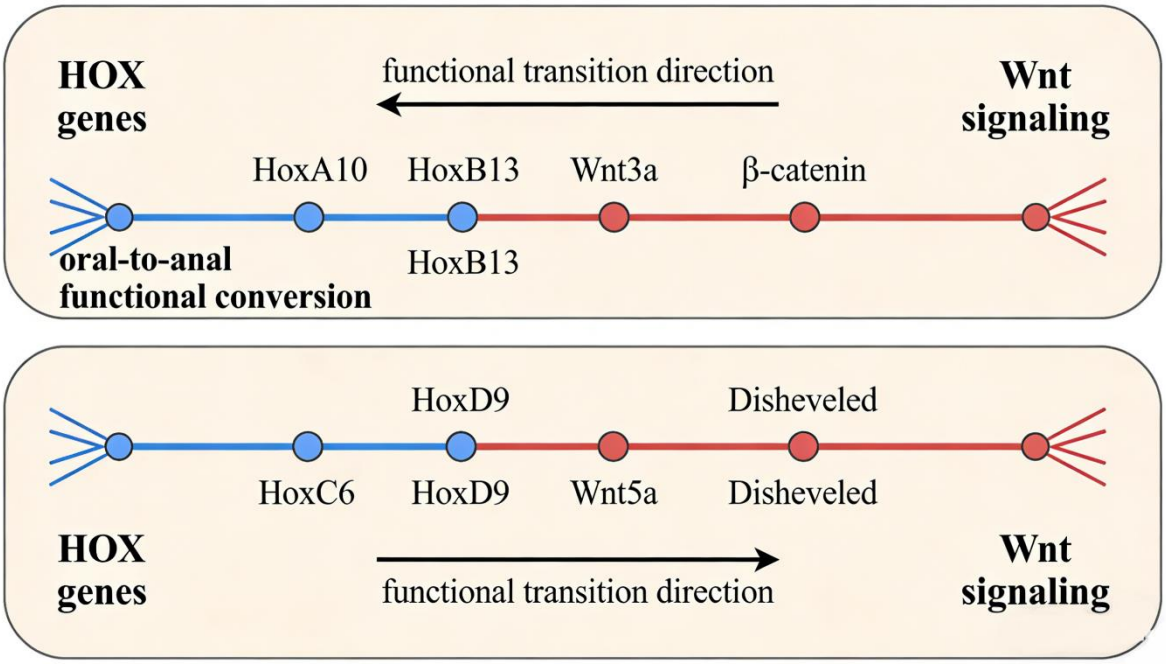

Figure 2. Molecular model of oral-anal functional exchange hypothesis
Proposed regulatory mechanism for bidirectional functional transformation between oral and anal mucosa.(A) Oral-to-anal functional conversion: Activation of HOX-A13 and Wnt3a signaling induces anal-type epithelial differentiation, sphincter remodeling, and acquisition of defecation control function.(B) Anal-to-oral functional conversion: Activation of HOX-D13 and inhibition of Wnt5a signaling promote salivary gland-like differentiation, mucosal secretion, and taste receptor expression.This model provides a mechanistic basis for organ functional compensation in end-stage digestive organ failure.

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